

Design and Implementation of a Mobile Robot System for Road Crack Detection

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Abstract—This paper presents Tank, a mobile robotic system for road crack detection that integrates a multi-degree-of-freedom manipulator with a tracked platform, offering flexible mobility in complex terrains. Equipped with an RGB-D camera, high-definition camera, LiDAR, industrial computer, and 5G module, the system forms a mobile inspection platform with real-time image acquisition and processing capabilities. Based on this, a road crack detection framework is proposed, covering image acquisition, dataset construction, deep learning model training and deployment, and crack detection. Combining self-collected images with the public CrackForest dataset trains Faster R-CNN and various YOLO models, and the best-performing model is deployed on Tank for real-world crack detection. Experimental results show that the system achieves an F1 of 0.8357 and a detection speed of 65.8 FPS, validating its practicality in road inspection tasks.

Keywords—tracked mobile robot, road crack detection, YOLOv11, autonomous road inspection

I. INTRODUCTION

The structural integrity of road infrastructure is crucial for the safe operation of transportation systems. Over time, pavement surfaces deteriorate due to traffic loads, material aging, and environmental factors such as temperature fluctuations and water infiltration, leading to various crack types [1]. Studies have shown that delayed detection of such cracks can accelerate pavement damage, increase accident risks, and raise maintenance costs [2]. As a result, automated and efficient crack detection has become a critical focus in pavement maintenance.

Recently, deep learning-based approaches—such as CNNs [3], FCNs [4], Faster R-CNN [5], and YOLO models [6]—have shown great promise in crack detection tasks [7]. However, their performance is often hindered by image quality variations and environmental disturbances like glare, shadows, or rain [8]. Most existing approaches rely on offline processing

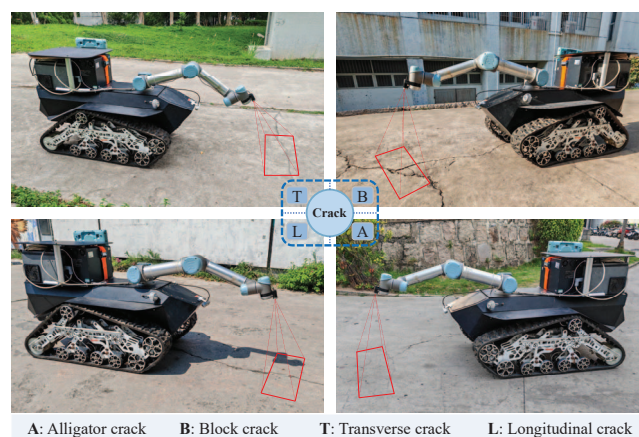


Fig. 1. The Tank captures and identifies various types of cracks under diverse scenarios.

and are rarely validated under dynamic, real-world conditions. Moreover, current systems commonly use wheeled robots or manual data collection, which limit efficiency and adaptability, particularly on unstructured terrains such as slopes, gravel roads, or grasslands [9].

Integrating deep learning with mobile robotics offers a promising solution. Compared with wheeled platforms, tracked platforms offer superior traction and terrain adaptability, especially in off-road or complex environments [10], [11], [12]. When combined with multi-degree-of-freedom manipulators, they enable flexible surface inspection and autonomous operation.

In recent years, mobile robots have made steady progress by integrating visual-inertial-LiDAR (VIL) techniques for localization and mapping. By fusing measurements from multiple sensors, these methods not only achieve high-precision self-localization in dynamic scenes but also support concurrent object tracking and three-dimensional environmental

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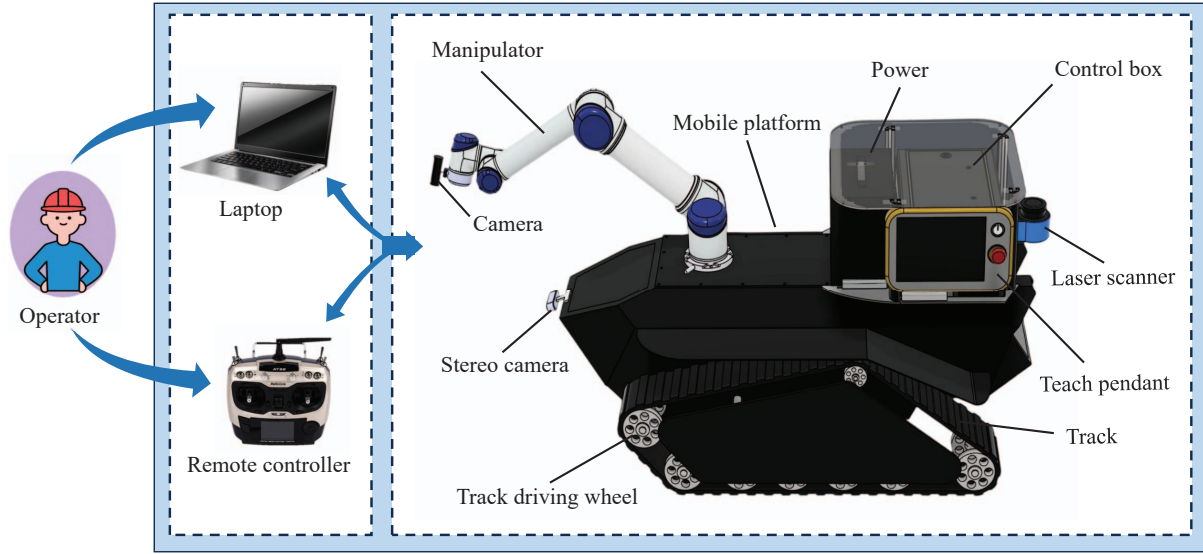


Fig. 2. Interaction between the operator and the Tank.

reconstruction [13], [14]. Building on this, lightweight deep feature extraction and matching networks further improve keypoint robustness and correspondence accuracy, enabling stable operation under fast motion, weak texture, and even partial occlusion [15].

Hence, in this paper, we present Tank, a tracked mobile robotic system designed for road crack detection. The system integrates a six-degree-of-freedom manipulator with a tracked chassis, enhanced by an industrial computer and multimodal sensors. As shown in Fig. 1, Tank can detect various crack types from different angles and distances, adapting to diverse scenarios. The key contributions of this study are as follows:

- 1) A tracked mobile robot is developed by integrating a multi-DOF manipulator, industrial computer, communication modules, and multimodal sensors on a tracked chassis. The system offers strong terrain adaptability and stable performance in complex environments.
- 2) A road crack detection framework is built upon the Tank platform, covering image acquisition, annotation, model training and evaluation, as well as deployment and field validation.
- 3) In real-world multi-scenario road environments, a YOLOv11-based crack detection model is deployed on Tank, achieving accurate and efficient crack identification.

II. HARDWARE AND SYSTEM DESIGN

This section outlines the hardware components, core subsystems, and control framework of the Tank robot. As illustrated in Fig. 2, Tank operates in remote-control mode, where the operator monitors live video via a terminal and uses a controller to manage movement, arm operations, and image capture for crack detection.

A. Tracked Mobile Platform

As shown in Fig. 2, the mobile platform adopts a tracked chassis structure, equipped with a 2960 Wh outdoor power supply mounted on top to ensure continuous operation of the entire robotic system. The platform features a hollow alloy frame, which achieves both lightweight construction and effective protection for the embedded industrial computer. Acting as the central control unit, the computer handles data processing, task scheduling, and state management for all components and subsystems. The drive system integrates high-performance servo motors with a customized tracked suspension, providing excellent maneuverability and terrain adaptability. These diverse mobility modes significantly enhance the platform's capability to navigate through various complex terrains.

Fig. 3 shows typical test scenarios used to assess the Tank platform's terrain adaptability. These include traversal over grass (a–b), gravel and uneven surfaces (c–d), speed bumps (e–f), and roadside curbs (g–h), demonstrating the robot's stability and obstacle-crossing capability across diverse environments.

B. Multi-Degree-of-Freedom Manipulator

In this work, the Tank is equipped with a UR5 six-degree-of-freedom manipulator to capture crack images. The manipulator features a repeatability of ± 0.1 mm and a maximum reach of 850 mm. A 1080p high-resolution camera is mounted at the end-effector to capture detailed visual data. The manipulator's articulated joint structure enables flexible pose adjustment, allowing it to adapt to varying road surface geometries and significantly enhance the accuracy and completeness of crack detection.

C. Control and Perception System

As shown in Fig. 4, a high-performance industrial computer (Intel Xeon, NVIDIA T1000) serves as the core component

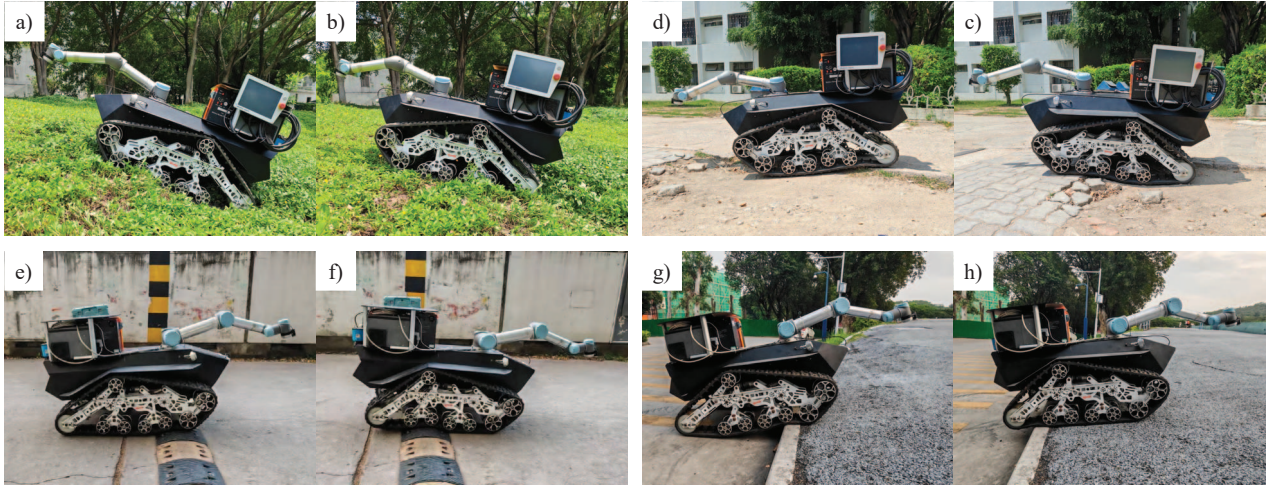


Fig. 3. Representative test environments for evaluating terrain adaptability of Tank.



Fig. 4. Subsystem hardware connections and data flow in the Tank.

of the tank, which can be used to manage remote inputs, motion control, sensor data processing, and crack detection in real time. To support comprehensive perception and enable future autonomous functions like SLAM [16] and navigation [17], the system incorporates a modular multi-sensor suite, including: (1) An Intel RealSense D455 stereo camera provides high-precision RGB-D data for near-field obstacle detection, road surface monitoring, and initial 3D crack profiling. (2) A 270° wide-angle 2D LiDAR delivers accurate distance measurements in the rear and rear-lateral areas. Combined with the front-facing camera, this setup enables nearly 360° environmental awareness. (3) A high-precision six-axis IMU continuously measures tri-axial acceleration, angular velocity, and orientation, supplying critical inputs for motion control, sensor fusion, and state estimation.

This flexible perception architecture can support remote operations, semi-autonomous or fully autonomous checks,

while enhancing the robustness and scalability of the system.

III. ROAD CRACK DETECTION FRAMEWORK BASED ON TANK

To validate the practical capabilities of the Tank platform, we present a road crack detection framework built upon it, as illustrated in Fig. 5. The workflow includes three main stages: 1) crack dataset construction, 2) model training and evaluation, and 3) field testing. The implementation details of each stage are outlined below.

A. Crack Dataset Construction

We employ the RGB-D camera and wireless communication module mounted on the Tank to capture and transmit real-time road surface images. Based on the visual feedback, the robot is remotely navigated to areas exhibiting potential cracks. The angle and distance of the manipulator's end-effector are then adjusted to enable close-range image acquisition using a 1080p high-definition camera. The captured crack images are transmitted in real time to the ground station for storage. Subsequently, all images are manually annotated to construct a dedicated dataset consisting of 777 images with a resolution of 640×480 pixels.

B. Model Training and Evaluation

To enhance the model's generalization and detection accuracy, the self-constructed dataset is combined with the public CrackForest Dataset [18]. A total of 716 images are used for training, 90 for validation, and 89 for testing.

Five mainstream object detection algorithms are selected for comparative evaluation: Faster R-CNN, YOLOv5, YOLOv7, YOLOv9, and YOLOv11 [19]. All models are implemented using the PyTorch framework with unified training parameters to ensure fair comparison. Experiments are conducted on a workstation equipped with an Intel® Xeon® Gold 6226R 2.90 GHz CPU and an NVIDIA GeForce RTX 3090 GPU (24 GB). All models are trained using the SGD optimizer [20] with a

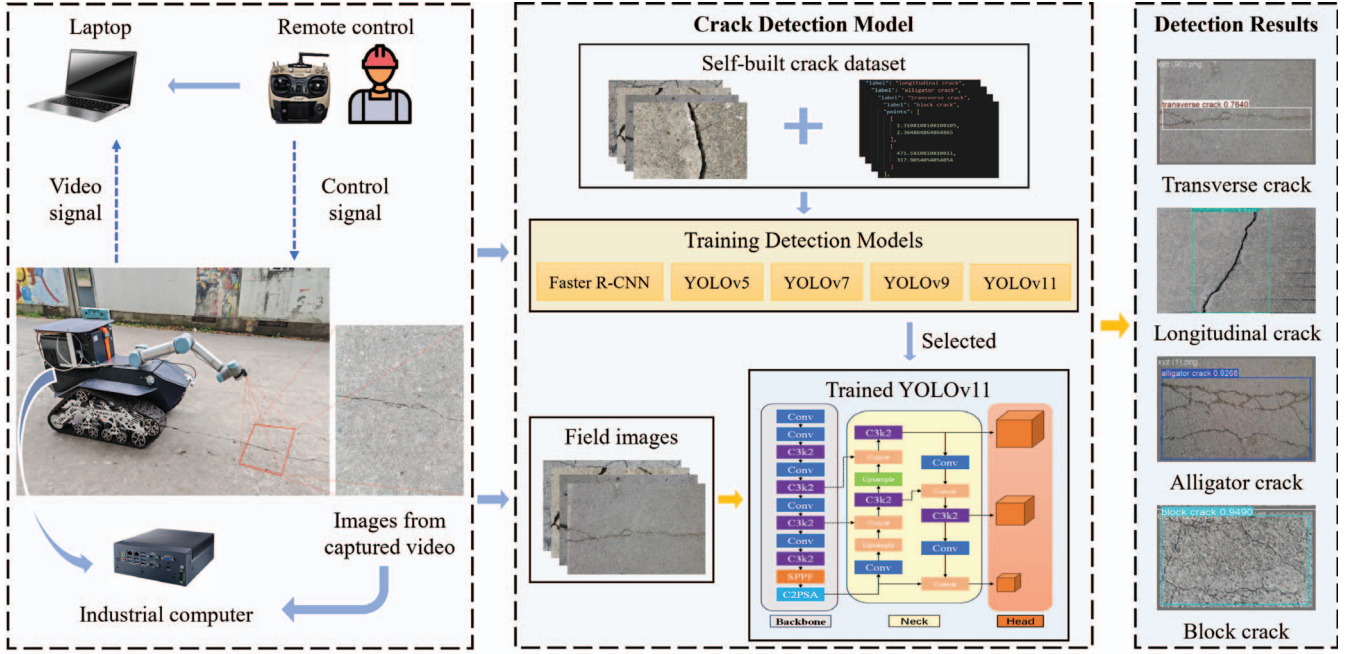


Fig. 5. Framework of road crack detection based on Tank.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT CRACK DETECTION MODELS

Model	Precision	Recall	F1-score	mAP@0.5	mAP@0.5:0.95	Parameters	FLOPs	FPS
Faster RCNN	0.4852	0.8262	0.6020	0.7017	0.3933	136.7M	324.8G	20.3
YOLOv5	0.8183	0.7990	0.8030	0.8247	0.5993	46.12M	107.7G	79.4
YOLOv7	0.7415	0.6661	0.7015	0.7150	0.4984	36.50M	103.2G	71.9
YOLOv9	0.7210	0.7623	0.7411	0.7708	0.5287	60.5M	263.9G	40.3
YOLOv11	0.8194	0.8579	0.8357	0.8853	0.6622	25.3M	86.6G	65.8

learning rate of $5e-3$, a batch size of 24, and 500 training epochs.

Eight evaluation metrics are used to assess the models' performance: Precision (Pr), Recall (Re), F1-score (F1), mAP@0.5, mAP@0.5:0.95, model parameters, FLOPs, and frames per second (FPS) [21]. Their calculation formulas are as follows:

$$Pr = \frac{TP}{TP + FP} \quad (1)$$

$$Re = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = 2 \cdot \frac{Pr \cdot Re}{Pr + Re} \quad (3)$$

$$mAP@[a:b] = \frac{1}{T} \sum_{k=1}^T \left(\frac{1}{4} \sum_{c=1}^4 AP_c(\tau_k) \right), \quad (4)$$

$$\tau_k \in \{a, a+0.05, \dots, b\}.$$

Here, TP , FP , and FN denote the numbers of true positives, false positives, and false negatives, respectively. The $F1$ score

is used to assess the overall performance of crack detection, where τ denotes the IoU (Intersection over Union) threshold, defined as the overlap ratio between predicted and ground-truth bounding boxes.

As shown in Table I, YOLOv11 outperforms all other models across most evaluation metrics. It achieves the highest Precision (0.8194), Recall (0.8579), and F1-score (0.8357), indicating its superior balance between accuracy and completeness in crack detection. Moreover, it obtains the best localization performance, with an mAP@0.5 of 0.8853 and an mAP@0.5:0.95 of 0.6622, surpassing all competitors. Additionally, YOLOv11 has the lowest parameters (25.3M) and the lowest FLOPs (86.6G), indicating high efficiency in both memory and computation. While YOLOv5 records the highest FPS (79.4), YOLOv11 still maintains real-time performance at 65.8 FPS, which is more than sufficient for practical deployment.

Fig. 6 presents a visual comparison of the detection results produced by different models on four common types of pavement cracks. Faster R-CNN tends to produce misaligned boxes, indicating a higher rate of false positives. YOLOv9 exhibits relatively low confidence in detecting transverse

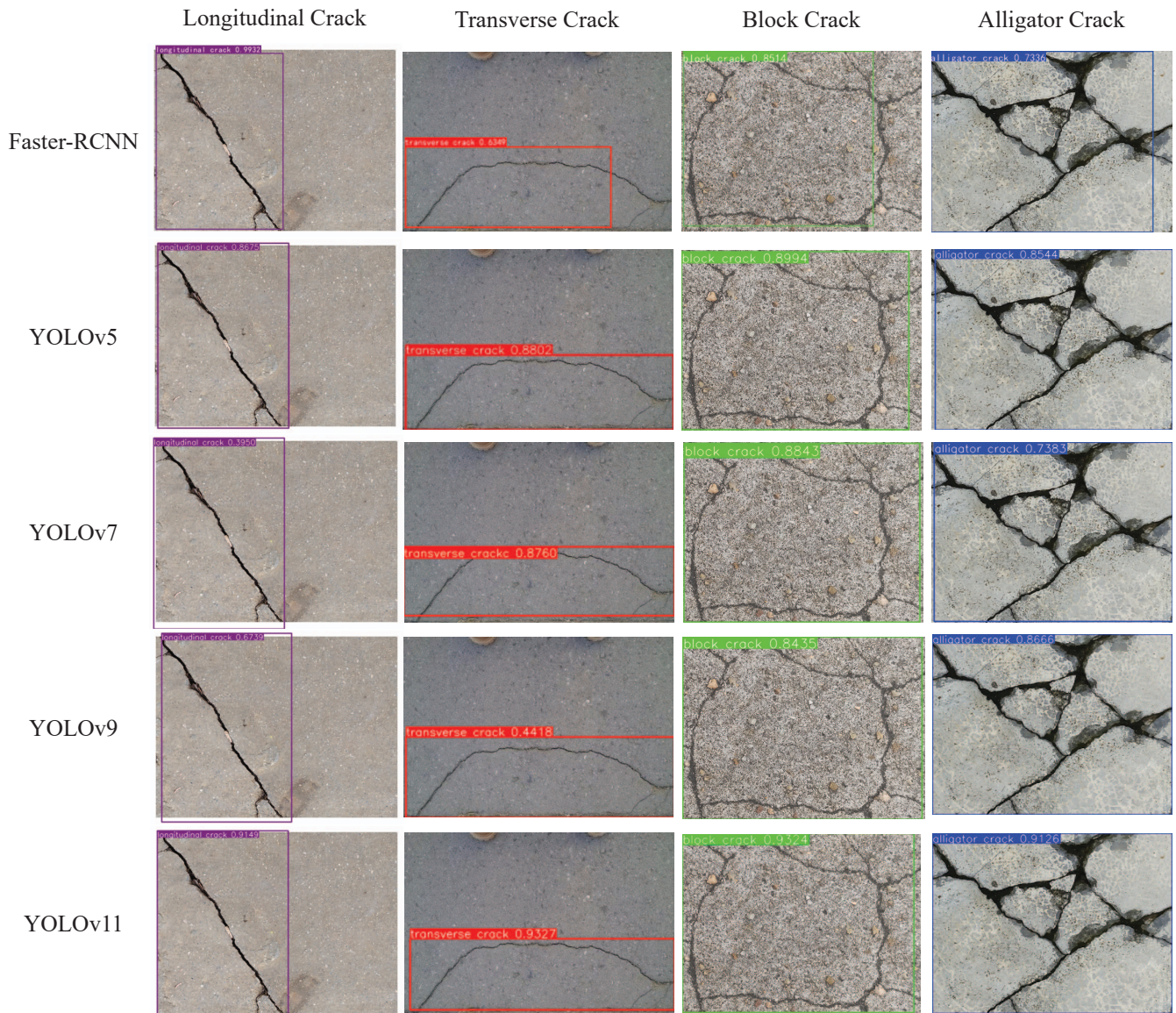


Fig. 6. Visual comparison of detection results produced by different models.

cracks, reflecting instability in certain scenarios. In contrast, YOLOv11 consistently delivers clearer and more accurate bounding boxes with higher confidence scores across all crack types, demonstrating strong robustness and completeness even in complex crack patterns.

C. Field Testing

To evaluate the practical applicability of the proposed system, the trained YOLOv11 is deployed on the Tank's industrial computer for field testing. As illustrated in Fig. 7, the model achieves accurate multi-target detection across various environments. It successfully identifies transverse and longitudinal cracks with confidence scores of 0.7556 and 0.7946, respectively. The model accurately delineates crack contours even under occlusions caused by leaves and branches, achieving

a confidence score of 0.7262. On textured surfaces, block cracks are detected with high confidence (0.8849), indicating strong discriminative capability. Despite challenging lighting conditions, including overexposure and low-light scenarios, the model maintained stable detection performance, with respective confidence scores of 0.6272 and 0.7947, underscoring its robustness to illumination changes.

IV. CONCLUSION

This paper presents the design and implementation of a tracked mobile robotic system, named Tank, which demonstrates excellent terrain adaptability and operational flexibility. Based on this platform, a comprehensive road crack detection framework was developed, integrating data acquisition, model training, and real-world deployment. Comparative experiments

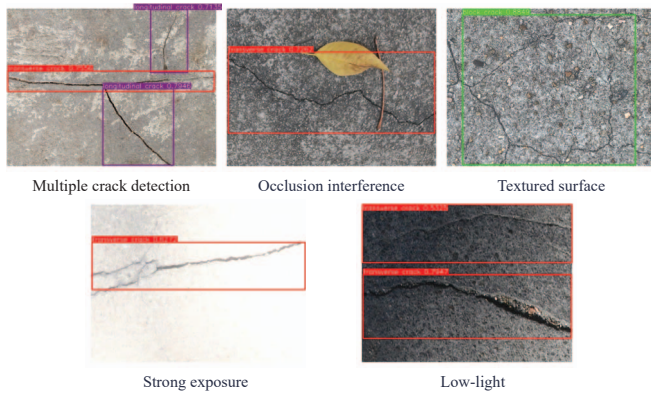


Fig. 7. Detection results of road cracks under real-world conditions.

with multiple state-of-the-art detection models confirmed that YOLOv11 delivers superior performance in both detection accuracy and real-time inference speed. Field tests further validated the system's robustness and practicality under diverse and challenging environmental conditions.

Future work will develop efficient, lightweight crack detection models tailored to embedded platforms to enhance the real-time recognition and energy efficiency of mobile robots under resource-constrained conditions; in parallel, we will implement a tightly coupled visual-inertial-LiDAR (VIL) SLAM system to exploit multimodal sensor complementarity and provide robust environmental perception and self-localization [22]. The integrated system will deliver accurate and continuous spatial information to support global path planning and local real-time obstacle avoidance. Ultimately, by deeply integrating the detection model with SLAM, we aim to establish a perception-localization-planning-execution closed loop that enables fully automated pavement crack inspection and autonomous operation.

REFERENCES

- [1] S. Zhang, Y. Cui, and W. Wei, "Low-temperature characteristics and microstructure of asphalt under complex aging conditions," *Construction and Building Materials*, vol. 303, p. 124408, 2021.
- [2] Z. Fan, C. Li, Y. Chen, J. Wei, G. Loprencipe, X. Chen, and P. Di Mascio, "Automatic crack detection on road pavements using encoder-decoder architecture," *Materials*, vol. 13, no. 13, p. 2960, 2020.
- [3] L. Alzubaidi, J. Zhang, A. J. Humaidi, A. Al-Dujaili, Y. Duan, O. Al-Shamma, J. Santamaría, M. A. Fadhel, M. Al-Amidie, and L. Farhan, "Review of deep learning: concepts, cnn architectures, challenges, applications, future directions," *Journal of big Data*, vol. 8, pp. 1–74, 2021.
- [4] J. Dai, Y. Li, K. He, and J. Sun, "R-fcn: Object detection via region-based fully convolutional networks," *Advances in neural information processing systems*, vol. 29, 2016.
- [5] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.
- [6] P. Jiang, D. Ergu, F. Liu, Y. Cai, and B. Ma, "A review of yolo algorithm developments," *Procedia computer science*, vol. 199, pp. 1066–1073, 2022.
- [7] G. Zhu, S.-L. Shen, J. Yao, M. Wang, J. Zhuang, and Z. Fan, "Automatic lightweight networks for real-time road crack detection with dpso," *Advanced Engineering Informatics*, vol. 68, p. 103610, 2025.
- [8] G. Zhu, J. Liu, Z. Fan, D. Yuan, P. Ma, M. Wang, W. Sheng, and K. C. Wang, "A lightweight encoder-decoder network for automatic pavement crack detection," *Computer-Aided Civil and Infrastructure Engineering*, vol. 39, no. 12, pp. 1743–1765, 2024.
- [9] G. Zhu, Z. Fan, P. Ma, W. Huang, Z. Ye, M. Huang, J. Li, Z. Jiang, Z. Zhong, and W. He, "Road crack acquisition and analysis system based on mobile robot and deep learning," in *2021 IEEE 11th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*, pp. 601–607, IEEE, 2021.
- [10] Y. You, Z. Fan, W. Chen, G. Zhu, B. Qiu, J. Xin, J. Chen, F. Deng, Y. Hou, W. Liang, *et al.*, "Design and implementation of mobile manipulator system," in *2019 IEEE 9th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*, pp. 113–118, IEEE, 2019.
- [11] C. Röhrig, D. Heß, and F. Künemund, "Motion controller design for a mecanum wheeled mobile manipulator," in *2017 IEEE conference on control technology and applications (CCTA)*, pp. 444–449, IEEE, 2017.
- [12] N. A. I. Ruslan, N. H. Amer, K. Hudha, Z. A. Kadir, S. A. F. M. Ishak, and S. M. F. S. Dardin, "Modelling and control strategies in path tracking control for autonomous tracked vehicles: A review of state of the art and challenges," *Journal of terramechanics*, vol. 105, pp. 67–79, 2023.
- [13] H. Zhang, D. Wang, and J. Huo, "A visual-inertial dynamic object tracking slam tightly coupled system," *IEEE Sensors Journal*, vol. 23, no. 17, pp. 19905–19917, 2023.
- [14] H. Zhang, D. Wang, and J. Huo, "Mounting misalignment and time offset self-calibration online optimization method for vehicular visual-inertial-wheel odometer system," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–13, 2024.
- [15] H. Zhang, J. Huo, Y. Huang, and Q. Liu, "Real-time dynamic visual-inertial slam and object tracking based on lightweight deep feature extraction matching," *IEEE Transactions on Instrumentation and Measurement*, 2025.
- [16] G. Grisetti, R. Kümmerle, C. Stachniss, and W. Burgard, "A tutorial on graph-based slam," *IEEE Intelligent Transportation Systems Magazine*, vol. 2, no. 4, pp. 31–43, 2011.
- [17] Y. Alkendi, L. Seneviratne, and Y. Zweiri, "State of the art in vision-based localization techniques for autonomous navigation systems," *IEEE Access*, vol. 9, pp. 76847–76874, 2021.
- [18] P. Li, M. Wang, Z. Fan, H. Huang, G. Zhu, and J. Zhuang, "Our-net: a multi-frequency network with octave max unpooling and octave convolution residual block for pavement crack segmentation," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [19] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," *arXiv preprint arXiv:2410.17725*, 2024.
- [20] S. Ruder, "An overview of gradient descent optimization algorithms," *arXiv preprint arXiv:1609.04747*, 2016.
- [21] G. Zhu, Z. Fan, J. Liu, D. Yuan, P. Ma, M. Wang, W. Sheng, and K. C. Wang, "Rha-net: An encoder-decoder network with residual blocks and hybrid attention mechanisms for pavement crack segmentation," *arXiv preprint arXiv:2207.14166*, 2022.
- [22] C. Zheng, W. Xu, Z. Zou, T. Hua, C. Yuan, D. He, B. Zhou, Z. Liu, J. Lin, F. Zhu, *et al.*, "Fast-livo2: Fast, direct lidar-inertial-visual odometry," *IEEE Transactions on Robotics*, 2024.